



TACAN Yardstick

Prepared by Stuka

NON CLASSIFIED
FOR PUBLIC RELEASE

PUBLISHED UNDER AUTHORITY OF THE 476TH vFG COMMAND STAFF



INTRODUCTION



CONTENT

This document explains the use of TACAN as yardstick to measure distance between two airframes.

TABLE OF CONTENTS

Table of Contents

Purpose 2

Operations 2

More Examples 4

 Example 1..... 4

 Example 2..... 5

 Example 3..... 6

Note 6



Purpose

TACAN yardstick can be used in (tactical) formations to measure the distance between 2 airframes. This helps the pilot to verify his position regarding to his leader.

Operation

Each TACAN channel uses two frequencies, one for transmit, and the other for receive.

If you separate the channels by 63, the receiver frequency of 1 airframe matches the transmitter frequency of another.

So for example:

Lead sets his TACAN to 11Y, and on AATR.



The wingman sets his to 74Y and also to AATR.



Once on the Navigation Mode Select Panel (NMSP), TCN is selected, the HSI will show you the distance in between the two airframes.



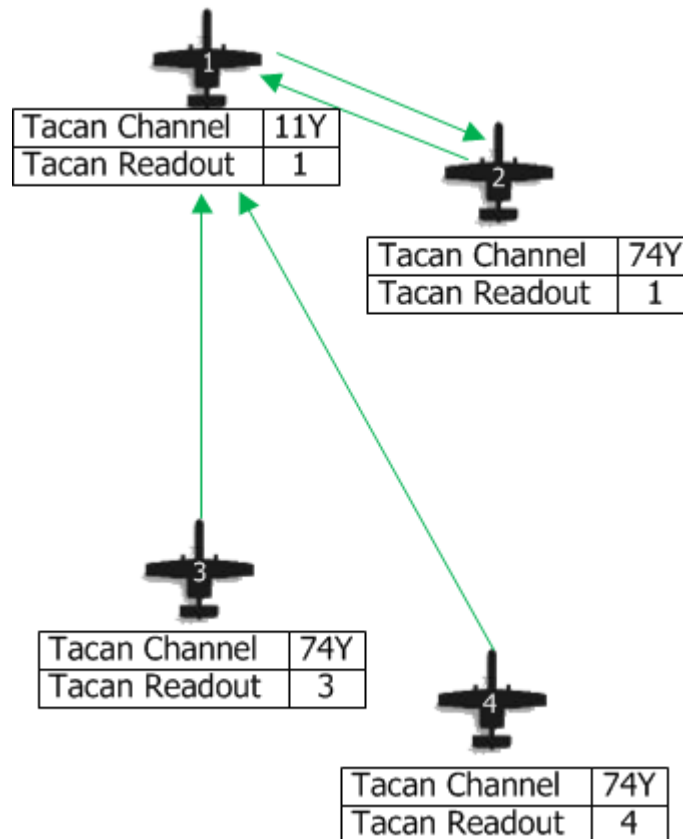
This distance is shown in both airframes.

In a flight with more ships, where the numbers 2,3 and 4 would set the same channel as above (74Y), the lead aircraft would get the reading towards to nearest airframe (strongest signal).



More Examples

Example 1



Lead set his TACAN to a certain channel, 11Y in this case.

Nr 2 and following, have to add 63 to that nr. That gives 74 (11+63) for nrs 2,3 and 4. Once they have setup their TACAN, all of them will get a distance readout on the HIS towards nr 1.

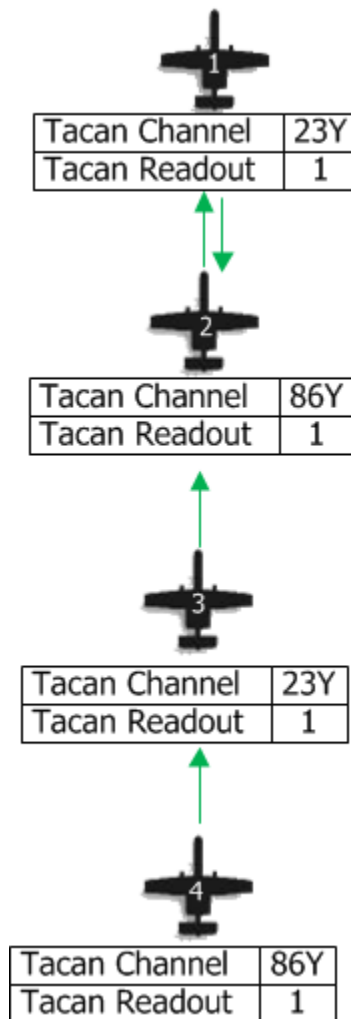
Nr 1 himself will get a readout to the closest emitter. So that's nr 2 in this example.

If nr 2 turns his TACAN of, 2 will obviously lose any TACAN readout.

Nr 1 his readout will switch to the distance to nr 3, because that's from that point the closest (and strongest) TACAN emitter for his configuration.



Example 2



Nr 1 has a setup of 23Y.

Nr 2 sets 86Y (23+63) and gets a readout of 1 mile towards nr 1. (nr 1 gets the same readout to nr 2 on his instruments)

Nr 3 can set the same as 2 (86), but in this example sets up his instrument towards nr 2. So for him that is again a difference of 63. TACAN goes only up to 126, so he has to subtract, giving him 23Y, the same as 1.

Nr 3 will get a distance readout towards nr 2 from here on.

A word of caution here: Since nr 2 is in between 2 emitters that match his configuration, he will get a readout towards the one that is the closest.

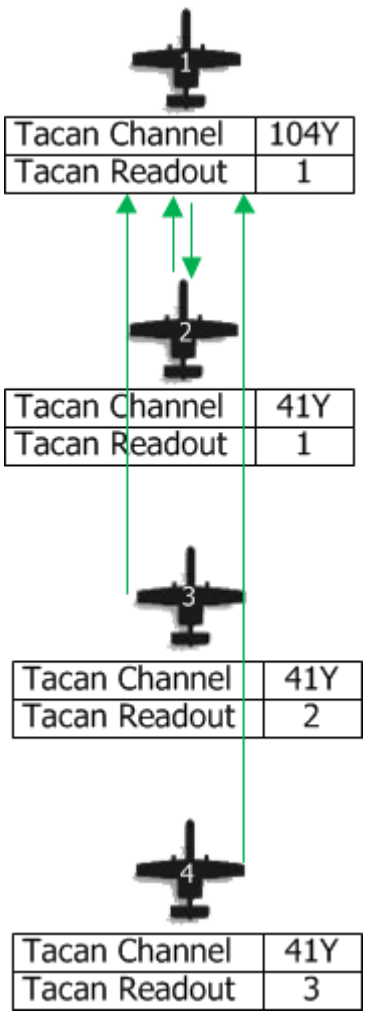
So as soon as Nr 3 is coming shorter towards him, then nr 2 is from 1, Nr 2 will get a readout towards Nr 3.

Nr 4 sets his TACAN instrument up referring towards nr 3's setup. That's 23+63, so 86Y again. He will get a distance readout towards nr 3.

From this point, the same warning goes out for nr 3 as earlier for nr 2. Nr 3 will be in between 2 matching emitters, and will have a readout towards the closest one.



Example 3



Here all aircraft are in trail again, and all have a setup towards nr 1.
Nr 1 has a setup of 104Y.
The others setup 41Y (104-63), and all get a readout towards nr 1.
Nr 1 himself will get a readout towards nr 2, since that is the closest emitter.

Note

Do not forget to turn TACAN off during FENCE IN.